Critical Appraisal of a Population-Based Case-Control Study on Early-Onset Colorectal Cancer in Ontario

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I. Introduction

The paper investigates risk factors for early-onset colorectal cancer, specifically is colorectal cancer diagnosed before age of 50. Rates of colorectal cancer are going up in younger adults, but have gone down in people 50 and older, likely because of screening [1]. This makes the topic important for both clinical care and public health. WHO's Global Health Estimates compare death and disability across age, sex, cause, country, and region and are described as useful for policy and resource allocation [3]. Statistics Canada's Health of Canadians also organizes population health using categories such as health determinants, chronic diseases, health behaviors, and mortality [4]. In this sense, the paper fits with this approach by looking at modifiable risk factors.

Scientific Rationale and Study Objective

The reasons for rising early-onset colorectal cancer are still not clear. Earlier studies have linked it to family history, obesity, alcohol use, inflammatory bowel disease, and diabetes. However, many recent studies did not include detailed lifestyle, diet information, or possible confounders. The authors also mention that exposures like Westernized diets, processed foods, antibiotics, allergies, and CT scans may be more common in younger people but have not been studied enough [1]. The study was set up to look at both previously established risk factors for overall colorectal cancer and a priori hypothesized novel factors. In this regard, the study was not purely exploratory, since they expected that both traditional and newer exposures could be linked to early-onset colorectal cancer.

This rationale fits with the idea of multicausality, where disease usually results from the joint action of several component causes rather than one single factor [2]. The study's goal was clear. The authors wanted to evaluate associations between a range of medical, lifestyle, and dietary factors linked to the risk of early-onset colorectal cancer [1].

Study Design

This was a population-based case-control study in Ontario, Canada. Cases came from the Ontario Cancer Registry. Controls were selected from the general population using random digit dialing [1]. Because people were chosen based on whether they had the outcome, this is a case-control study, not a cohort or cross-sectional study. That is why the authors reported odds ratios instead of risks or incidence rates. By comparing people with early-onset colorectal cancer to controls of the same age, the study could look at many exposures for a rare outcome.

II. Methods

Data Sources and Study Populations

Cases were identified through the Ontario Cancer Registry, a population-based cancer registry [1]. Eligible cases were Ontario residents aged 20 to 49 years with pathologically confirmed incident invasive colorectal adenocarcinoma diagnosed between January 2018 and May 2019, and the paper also reports the ICD-O-3 topography codes used for case identification [1]. This is a strength because case ascertainment was based on registry and pathology information rather than self-report. The authors also used pathology report data for some tumor characteristics. For example, microsatellite instability status was classified using immunohistochemistry staining results for MLH1, MSH2, MSH6, and PMS2 [1]. Overall, this makes the case definition specific

and likely reduces outcome misclassification. Controls were recruited separately in 2019 using modified random digit dialing. They had to be Ontario residents aged 20 to 49 with no history of colorectal cancer. Controls were matched to cases by sex and 5-year age group.

There were no data linkages across multiple administrative databases. The study mainly used the cancer registry for cases and the questionnaire for exposures. The data sources and methods were described clearly, but outcome measurement was stronger than exposure measurement.

The eligibility criteria were clear, but this does not guarantee that cases and controls are fully comparable. The case definition was specific and confirmed by pathology. However, controls had to be reached by phone, agree to take part, and be able to do the online survey. This recruitment process could have introduced differences between participants and the source population.

The selection process itself was described clearly, but it also raises some concerns. Cases were identified from the Ontario Cancer Registry, which is a complete source for the province [1]. This means the cases were population-based, not just from one hospital. In a case-control study, controls are supposed to reflect the exposure distribution in the population that produced the cases. Recruiting controls from the general population by random digit dialing was an attempt to do this, although low participation means this may not have been fully achieved.

Participation was selective at several stages. Out of 782 eligible cases, 220 agreed to be contacted and 175 finished the questionnaire. For controls, about 53,000 households were called, 1,800 eligible people were found, 640 showed interest, and 253 completed the questionnaire [1]. This could lead to selection bias, especially for controls.

Missing data was handled simply and transparently. The authors made unknown categories for variables with more than 5% missing data. These are shown in the tables for variables like income, CT scans, and secondhand smoke exposure. The paper did not have other advanced methods [1].

Variables

The outcome was early-onset colorectal cancer, with cases defined as Ontario residents aged 20 to 49 years with pathologically confirmed incident invasive colorectal adenocarcinoma. For some case characteristics, the paper also defined tumor subsite and microsatellite instability status using pathology-based information [1].

Exposure and covariate data were collected using a self-administered web-based questionnaire. The questionnaire asked about sociodemographic factors, family history of colorectal cancer, personal medical history, medical procedures, medication use, female reproductive history, BMI, smoking, secondhand smoke, alcohol use, physical activity, sedentary time, supplement use, and many dietary variables. Most lifestyle factors were reported 2 years ago. Most diagnoses, procedures, medications, and supplements were limited to events at least 2 years before the questionnaire [1]. This was done to measure exposures from before the cancer diagnosis, not after symptoms started. The goal was to reduce reverse causation and make the timing of exposures more relevant. Even so, exposure measurement relied on self-report, which is more likely to have recall bias and measurement error than registry-based case ascertainment. The paper also defines variables like family history, medication use, parity, menopausal status, and tumor subsite in the table footnotes [1].

BMI was calculated as weight in kilograms divided by height in meters squared. Physical activity came from a modified Godin-Shephard questionnaire. Sedentary time was based on average sitting time during the week and weekends. Dietary variables were measured by how often certain foods were eaten for specified serving sizes. The authors also made a Western-like dietary pattern score using six non-beneficial and three beneficial dietary components, with higher scores meaning a more Westernized diet [1].

Methods to Control for Confounding

Potential confounders were described clearly. The models included potential confounders such as age, sex, family history, aspirin or NSAID use, smoking, physical activity, BMI, alcohol, red and processed meat, fruit and vegetables, high-fiber foods, and calcium supplements. Diabetes, inflammatory bowel disease, race, education, and income were also checked as confounders and kept if removing them changed the odds ratio by at least 15% [1].

The main analysis used binary logistic regression to estimate odds ratios adjusted for age and sex, and then multivariable-adjusted odds ratios with 95% confidence intervals. Each model included the main exposure and a set of covariates chosen because they are known risk or protective factors. The authors also looked at effect modification by sex and by tumor subsite. This means the analyses were done by sex, tested for interaction with likelihood ratio test, and used polytomous logistic regression by tumor subsite [1]. This let them see if associations were different in subgroups.

Potential Sources of Bias

Participation was a concern. Response rates were low, especially for controls, and this could cause response bias. The authors compared participating cases to eligible non-participating cases

from the Ontario Cancer Registry and found that participants were more likely to be female, but similar in age, stage, and tumor subsite. Controls had smoking and obesity rates similar to the general young adult population in Ontario. Moreover, survivor bias was probably limited because most cases were recruited within 10 months of diagnosis, which is not long after diagnosis.

How exposures were measured is another issue. The study used self-reported questionnaires, so recall bias is possible. The authors tried to reduce this by asking about diet and lifestyle two years ago and by limiting most diagnoses, procedures, medications, and supplements to events at least two years before the questionnaire. This was to focus on exposures before diagnosis. The paper also mentions reverse causality, where early symptoms or weight loss could explain the BMI and antacid findings. Some results may also be false positives because many variables were tested.

Measurement error is also important. Key exposures like diet, sedentary time, physical activity, and past medication use are hard to measure accurately with self-report. So, some variables may have been misclassified. Residual confounding is still possible, since not all factors may have been fully measured. The paper is aware of these main biases, but most were handled by design and discussion, not fully removed by analysis.

III. Results

Participant Characteristics

Table 1 shows the characteristics of the 175 cases and 253 controls. The mean age was 43 for cases and 40 for controls. Most were female, white, had at least a college degree, and lived mainly in urban areas. The table also lists country of birth, education, income, occupation, and for cases, tumor subsite, microsatellite instability status, and stage at diagnosis. Among cases with data, 24% had proximal colon cancer, 28% had distal colon cancer, 48% had rectal cancer, 11% had MSI-high tumors, and 60% were diagnosed at stage III or IV [1].

Unadjusted and Adjusted Estimates

The results tables show odds ratios adjusted for age and sex, and multivariable-adjusted odds ratios. For example, family history of colorectal cancer had an age- and sex-adjusted OR of 2.20 and a multivariable-adjusted OR of 2.37. Allergy or asthma had an adjusted OR of 0.68 and a multivariable-adjusted OR of 0.62 [1]. The paper also gives adjusted results for lifestyle and diet, like sedentary time, sugary drinks, Western-like diet score, and calcium supplements. The main tables do not show crude odds ratios, only adjusted ones.

Measure of Precision

All the main estimates were reported with 95% confidence intervals. For example, family history of colorectal cancer had an adjusted OR of 2.37 with a 95% confidence interval of 1.47 to 3.84, sedentary time of at least 10 hours per day had an adjusted OR of 1.93 with a 95% confidence interval of 1.02 to 3.65, and sugary drinks of at least 7 per week had an adjusted OR of 2.99 with a 95% confidence interval of 1.57 to 5.68 [1].

IV. Discussion

One issue is the low response rate, especially among controls. The authors say this may have introduced response bias. They note that the controls were similar to the general young adult population in Ontario for smoking and obesity. For the cases, participation was more common among females. However, participants and non-participants were still similar in age, stage, and tumor subsite. The authors also discuss survivor bias. They suggest it was probably limited because most cases were recruited within a few months of diagnosis [1].

Another concern is how the exposure data were collected. The study used questionnaires, so recall bias is possible, especially for things like childhood antibiotic use. The authors also note that reverse causality may have affected some findings, such as BMI and antacid use. Because the study looked at many different variables, some of the reported associations may simply be false positives [1].

The discussion connects back to the main purpose of the study. The authors summarize the main associations they found. They suggest that modifiable factors, especially sedentary behavior and unhealthy diet including sugary drinks, may be related to early-onset colorectal cancer risk. If these findings are replicated, they may help guide prevention strategies for younger people [1].

Overall, this was a useful population-based case-control study on an important public health issue. A major strength is that cases were identified through a cancer registry. This makes disease definition clearer and more reliable. The study also looked at a wide range of exposures and reported adjusted odds ratios with confidence intervals [1]. At the same time, the findings need to be interpreted carefully. Selection bias, recall bias, measurement error, and residual confounding are all possible [1,5]. For these reasons, observed associations do not automatically prove

causation [5]. The paper is more convincing as evidence of important associations than as proof of clear causal effects. Even so, it adds useful public health knowledge by pointing to modifiable factors that may matter for prevention in younger adults [1,3,4].

Reference

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